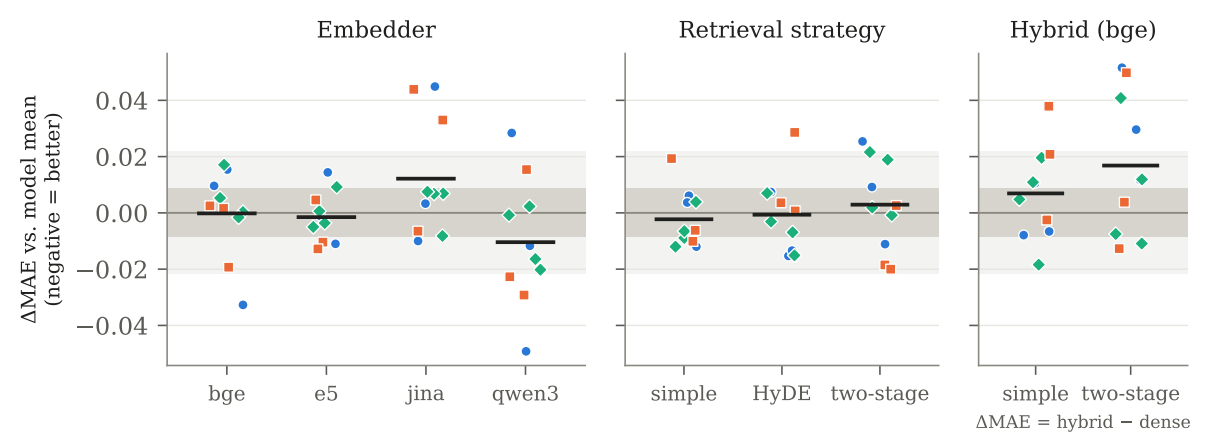
Embedder
Retrieval strategy
Hybrid (bge)
ΔMAE vs. model mean
(negative = better)
0.04
0.02
0.00
−0.02
−0.04
bge
e5
jina
qwen3
simple
HyDE
two-stage
simple
two-stage
ΔMAE = hybrid − dense
Llama
Ministral
Qwen
mean over models
± run-to-run SD, same chunks (0.009)
± run-to-run SD, HyDE reruns (0.022)